Generative Models for Cat Image Synthesis

Deep Learning: Project III

Warsaw University of Technology

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Goal & Motivation

Train and compare three generative architectures on cat images at 64×64 resolution with a 6 GiB VRAM constraint.

Three approaches

1. **DCGAN** with Wasserstein loss and gradient penalty (WGAN-GP)
2. **VQ-VAE** with autoregressive PixelCNN prior
3. **DDPM** with DDIM fast sampling

Evaluation: Fréchet Inception Distance (FID) on 5 000 samples and visual inspection of generated images.

Real cat samples (64×64)

Dataset & Preprocessing

CAT Dataset (Kaggle)

- 9 997 RGB cat photos, varying resolutions
- Subdirectories CAT_00 to CAT_06

Preprocessing

1. Resize shorter edge to 64 px
2. CenterCrop 64×64
3. Normalize to $[-1, 1]$

Fixed seed 42. No augmentation in baselines. Augmentation tested in ablation: FID 284.05 ± 14.80, worse than baseline 264.25 ± 20.21.

DCGAN with WGAN-GP

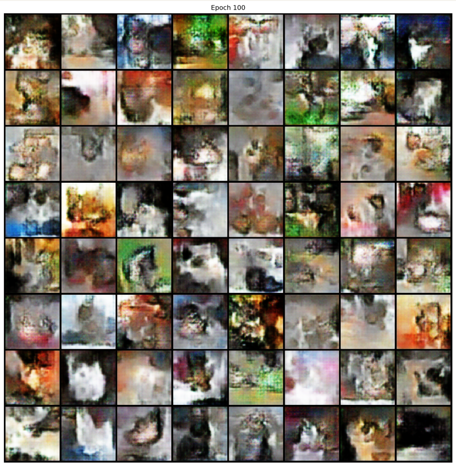
Architecture

- Generator: $\mathbf{z} \in \mathbb{R}^{128}$, 4 transposed-conv blocks, Tanh
- Discriminator: 4 strided-conv (LeakyReLU), scalar logit

Training

- WGAN-GP: $\lambda_{GP} = 10$, $n_{\text{critic}} = 5$
- Adam $\beta_1 = 0.5$, $\text{lr} = 2 \times 10^{-4}$
- 100 epochs, batch 64

Result: FID 221.51 — as shown later, the main bottleneck was an insufficient number of generator updates.



Epoch 100

VQ-VAE with PixelCNN Prior

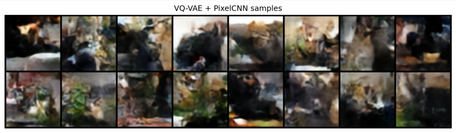
Phase 1: Autoencoder (100 epochs)

- 64×64 to 16×16 discrete code map
- Codebook: $K = 512$, $d = 64$, EMA updates ($\gamma = 0.99$)
- Commitment cost $\beta = 0.25$

Phase 2: PixelCNN Prior (100 epochs)

- 15-layer masked residual convolutions, 128 filters
- Trained on 256-token sequences (frozen encoder)
- Codebook perplexity: 403/512 (79% utilisation)

Result: FID 187.54 — autoencoder reconstructs well (MSE 0.0066), but PixelCNN prior limits sample quality. Early run with $K = 1024$ had codebook collapse (30% utilisation); fixed by switching to EMA updates and $K = 512$.



PixelCNN prior samples (low visual quality)

DDPM with DDIM Sampling

Architecture

- U-Net, channel mults (1, 2, 2, 2), 2 res-blocks per level
- Self-attention at 16×16; EMA weights at inference

Diffusion process

- Linear schedule: $T = 1000$, $\beta_1 = 10^{-4}$, $\beta_T = 0.02$
- DDIM ($\eta = 0$): 50 steps gives $20\times$ speedup
- 200 epochs, batch 16 (6 GiB VRAM limit)

Result: FID 24.11 (EMA weights, 50 DDIM steps) — best result by a wide margin. EMA alone drops FID 57% vs. non-EMA at the same step count.

Epoch 200 (EMA weights)

FID Comparison

Fréchet Inception Distance (5 000 samples)

Model	FID ↓	Training
DCGAN (WGAN-GP, n_c=5)	221.51	∼2 h
↪ corrected (std, n_c=1)	98.99	∼3 h
VQ-VAE+PixelCNN	187.54	∼4 h
DDPM+EMA	**24.11**	∼12 h

Not compute-normalised; shows best feasible config per model. All single-run. Original DCGAN was under-trained (see §DCGAN Fix).

Why such a large gap?

- DDPM has a well-defined MSE objective: smooth, stable gradients
- EMA weights reduce FID by 57% vs. non-EMA at 50 steps
- GAN mode collapse limits diversity despite WGAN-GP
- VQ-VAE: good reconstruction, PixelCNN prior limits sample quality

Consistent with *Diffusion Models Beat GANs* (Dhariwal & Nichol, 2021)

Generated Samples: Side by Side

(a) DCGAN
FID 221.51

(b) VQ-VAE+PixelCNN
FID 187.54

(c) **DDPM**
FID 24.11

DCGAN: artifacts, limited diversity | VQ-VAE: structure largely indistinguishable | **DDPM: clearly recognisable cats**

VQ-VAE Reconstruction Quality

Top: input | Bottom: VQ-VAE reconstruction (epoch 100)
MSE = 0.0066 | VQ loss = 0.0033 | perplexity 403/512 (79%)

DDPM: DDIM Steps and EMA Effect

FID vs. DDIM steps (non-EMA)

Steps	FID ↓	Speedup
10	66.90	100×
25	60.26	40×
50	**55.43**	**20×**
100	48.35	10×
200	39.56	5×

EMA at 50 steps: −57% FID

Weights	FID ↓
Without EMA	55.43
With EMA	**24.11**

50 steps: practical sweet spot (20× speedup).
EMA more impactful than doubling the steps.
Below 25 steps: FID degrades by 37% vs. 100 steps.

DCGAN: Latent Dimension and Learning Rate

z_{dim} ablation (3 seeds)

z_{dim}	FID mean±std	Δ
64	246.65 ± 5.57	−17.60
128 (baseline)	264.25 ± 20.21	-
256	266.05 ± 2.52	+1.80

z_{64} best, but within baseline σ. High baseline variance (±20) shows seed sensitivity.

Learning rate ablation (3 seeds)

LR	FID mean±std	Δ
1×10^{-4}	295.11 ± 18.27	+30.86
2×10^{-4} (base)	264.25 ± 20.21	-
4×10^{-4}	**220.05 ± 4.47**	−44.20

LR has far more impact than z_{dim}.
Low LR: generator cannot keep up with the critic.

Does Training on Dogs Help Cats?

Setup

- 12 500 cats + 12 500 dogs (Kaggle)
- Same DCGAN, 100 epochs, unconditional
- FID measured separately vs. cats and dogs

Results

Model	vs. Cats	vs. Dogs	vs. Combined
Cats only	221.51	-	-
Cats+Dogs	206.57	197.79	**190.39**

Combined FID (2 500 cats + 2 500 dogs) more appropriate for a mixed-class model. 15-pt cat improvement is within seed variance (±20) — suggestive, not conclusive.

Qualitative: visual quality poor — most samples are indistinguishable blobs. Training FID worsened across epochs, suggesting the mixed dataset destabilised the generator.

Cats+Dogs DCGAN (epoch 100)

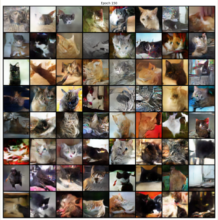
DCGAN: Why FID Was So High and How It Was Fixed

Post-hoc diagnosis: generator update budget

- $n_{\text{critic}} = 5$: 5 D-updates per G-update
- 156 batches/epoch $\Rightarrow$ only **3 100 G-steps** in 100 ep.
- Standard DCGAN ($n_c = 1$): **15 600** at same cost

Two corrected configs

- **Standard DCGAN**: switch to BCE loss; $n_c = 1$ is the correct standard (not a compromise), $g_{\text{ch}} = 128$, 150 ep. $\Rightarrow$ **23 400 G-steps**
- **WGAN-GP** $n_c = 2$: pragmatic fix; fewer critic steps may leave the critic less well-optimised — compute-equivalent n_c=5 would need ≈500 ep., 200 ep. $\Rightarrow$ **15 600 G-steps**



Std. DCGAN (epoch 150) — better visual quality

WGAN-GP $n_c = 2$ (epoch 200) — visually worse

FID results

Config	G-steps	FID
Original	3 100	221.51
Std. DCGAN	23 400	**98.99**
WGAN-GP $n_c = 2$	15 600	158.48

Research Questions: Answers

RQ1: Architecture trade-off under 6 GiB VRAM

DDPM: best quality (FID 24.11, 12 h, 5.5 GiB). DCGAN: fastest (2 h, $<$2 GiB) but mode collapse. VQ-VAE: stable training, PixelCNN prior is the bottleneck.

RQ2: Does adding dogs improve cat generation?

Possibly, but not conclusively. Cat FID improves from 221.51 to 206.57, but the 15-pt gap is within the observed DCGAN seed variance ($\pm$20). Class coverage and absence of hybrid artefacts would require a classifier-based analysis.

RQ3: Effect of latent dimension $z_{\text{dim}} \in \{64, 128, 256\}$

Differences are within baseline seed variance ($\pm$20); no firm conclusion about optimal z_{dim} can be drawn. $z = 64$ best on mean FID but not conclusively.

RQ4: DDIM speed-quality trade-off

50 steps: 20$\times$ speedup, FID 55.43 (non-EMA) or 24.11 (with EMA). Below 25 steps quality degrades sharply. 50 steps is the practical sweet spot.

Conclusion

Summary

Model	FID	Strength
DDPM	**24.11**	quality
VQ-VAE	187.54	stability
DCGAN	221.51	speed

- In our setup, DDPM clearly outperforms GAN and VAE
- For DDPM, EMA more impactful than doubling DDIM steps
- DDIM enables practical use at 50 steps
- Dog images: suggestive improvement, not conclusive
- DCGAN bottleneck fixed: $7\times$ more G-updates $\rightarrow$ FID 98.99

Main challenges

- 6 GiB VRAM: DDPM batch size limited to 16
- DCGAN mode collapse: WGAN-GP (label smoothing inactive under GP path)
- Codebook collapse at $K = 1024$: fixed with EMA, $K = 512$
- 8 h inference: solved with DDIM (50 steps, 25 min)
- DCGAN $n_{\text{critic}} = 5$: generator starvation fixed

Future work

- DDPM at 128×128 with gradient checkpointing
- Conditional DCGAN on cats+dogs
- Transformer prior (VQ-VAE-2 style)
- Cosine noise schedule

Thank you.

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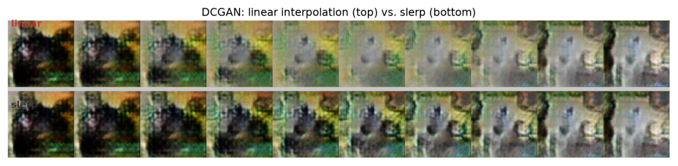
Latent Space Interpolation: Linear vs. Slerp

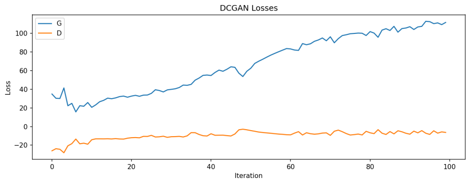


DCGAN — **linear** (top) vs. slerp (bottom)

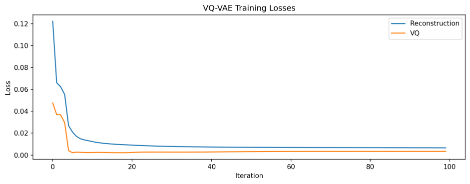
DDPM — **linear** (top) vs. slerp (bottom)

Linear interpolation is the primary method (top rows). Slerp is shown for comparison: it preserves the norm of the latent vector, avoiding the low-density region near the origin of $\mathcal{N}(0, I)$ that causes blurry midpoints in linear interpolation (most visible in DDPM).

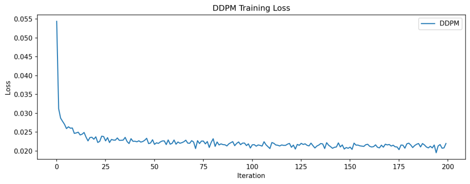
Training Dynamics



DCGAN (G vs. D) — adversarial oscillation



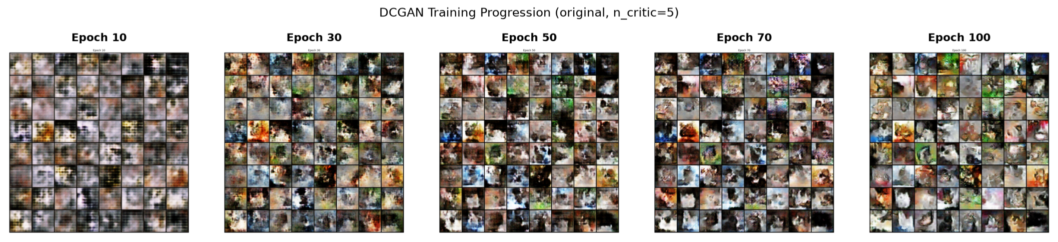
VQ-VAE (recon + VQ) — steady decrease



DDPM (ℓ_2 noise) — smooth convergence, most stable

DDPM's MSE objective gives the most stable training. DCGAN shows adversarial oscillation; VQ-VAE converges reliably with EMA codebook.

DCGAN Training Progression



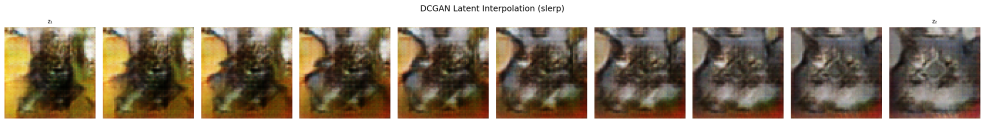
From blurry noise to recognisable cats — but likely limited by low generator update budget (3 100 G-steps at n_c=5).

Experimental Grid: What We Tested

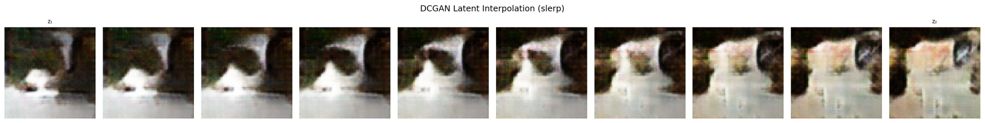
Model	Ablation / variant	Values tested	Metric
DCGAN	Latent dim z_{dim}	64, **128**, 256	FID, slerp smoothness
	Learning rate	1e-4, **2e-4**, 4e-4	FID
	Augmentation	off (**base**), on	FID
	Training data	cats only, **cats+dogs**	FID vs cats/dogs
DDPM	DDIM steps	10, 25, **50**, 100, 200	FID, wall time
	EMA weights	off, **on** (decay 0.9999)	FID
VQ-VAE	Codebook size K	1024 (collapsed) $\to$ **512**	perplexity, FID
Post-hoc fix	Standard DCGAN ($n_c = 1$)	150 ep, $g_{\text{ch}} = 128$	FID
	WGAN-GP $n_c = 2$	200 ep	FID

Bold = chosen baseline. Primary metric: FID on 5 000 generated vs 5 000 real images. Secondary: latent interpolation smoothness, reconstruction MSE (VQ-VAE).

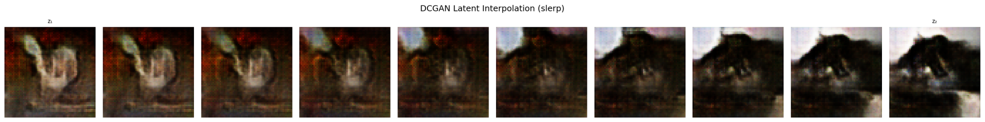
Latent Space Smoothness vs. z_{dim}



$z_{\text{dim}} = 64$



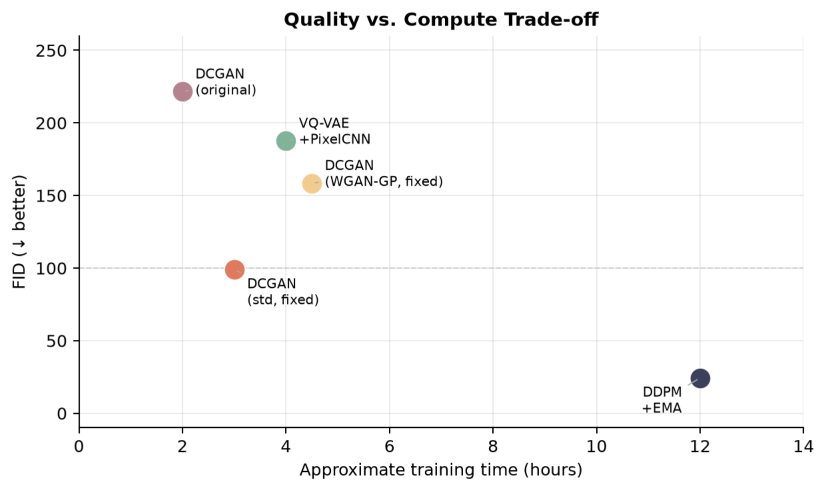
$z_{\text{dim}} = 128$ (baseline)



$z_{\text{dim}} = 256$

Slerp, 10 frames. Transitions smooth for all variants. z_{64} slightly softer; z_{128} and z_{256} indistinguishable.

Quality vs. Compute Trade-off



DDPM dominates in quality; DCGAN fastest. Fixed DCGAN closes the gap significantly.